SKILL 1

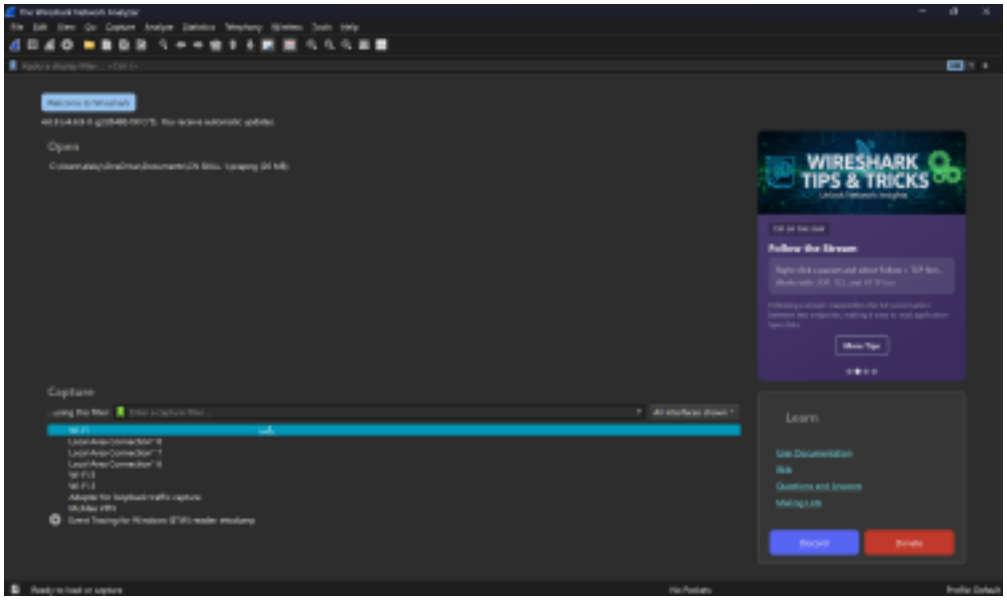
ENCAPSULATION BUILDER – WIRESHARK OBSERVATION RECORD

Aim: To capture network packets and understand how Ethernet, IP and TCP/UDP headers are added during data transmission.

Method: The values below are read from the packet list and packet-bytes/details panes in the supplied screenshots. Where a field is not expanded, the value is calculated from the visible packet bytes and is identified accordingly.

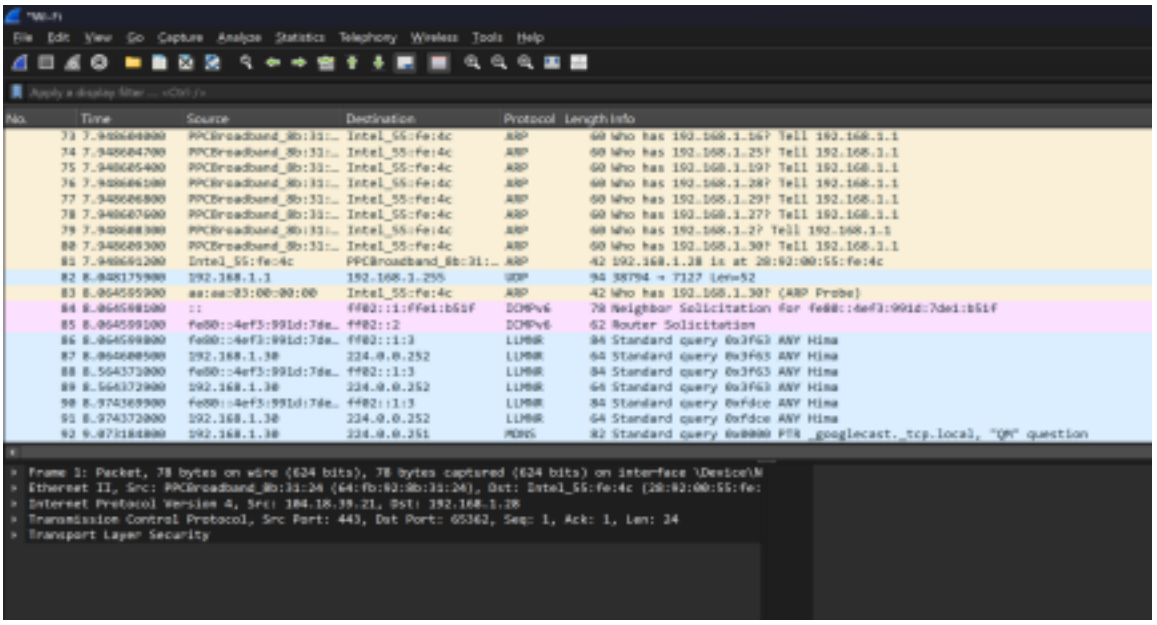
Step 1: Open Wireshark and select the interface

Wireshark is installed and opened. The welcome screen shows Wi-Fi selected in the available capture-interface list.



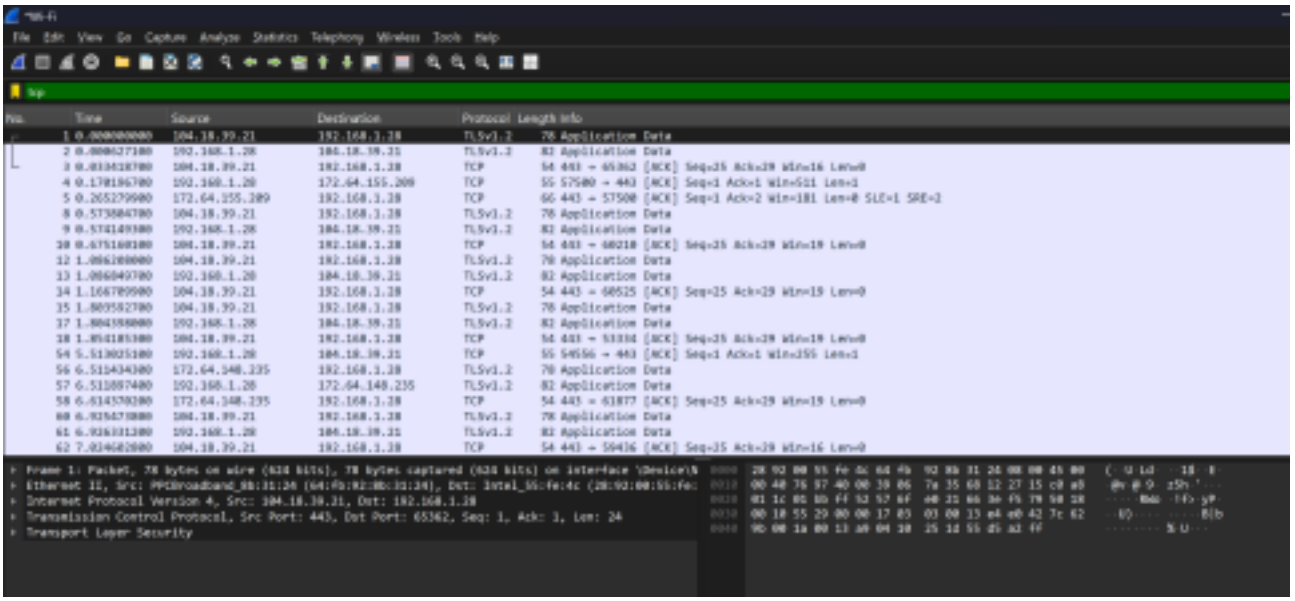
Step 2: Capture network packets

The capture is running on the Wi-Fi interface. The packet list contains ARP, UDP, ICMPv6, LLMNR and mDNS traffic. The selected packet details pane shows the protocol layers.



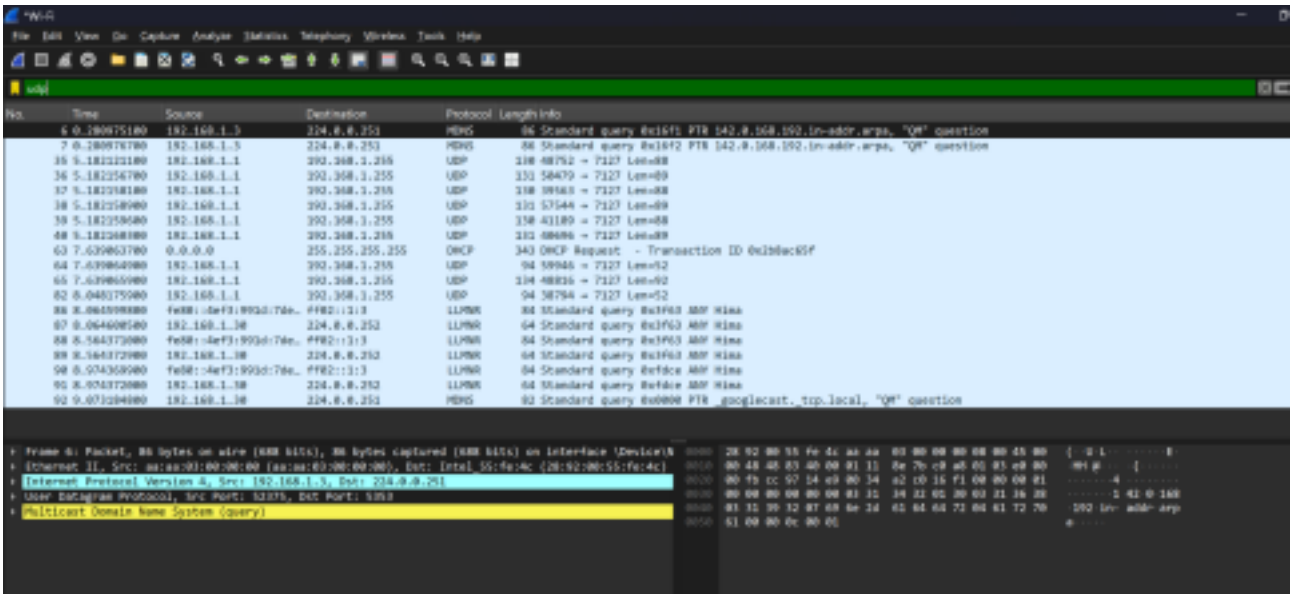
Step 3: Apply the TCP display filter

The display filter tcp is applied. The packet list shows TCP and TLS-related traffic. The selected packet is Frame 1, with 78 bytes captured. Its visible packet details identify IPv4 source 104.18.39.21, destination 192.168.1.28, TCP source port 443, destination port 65362 and TCP payload length 24 bytes.



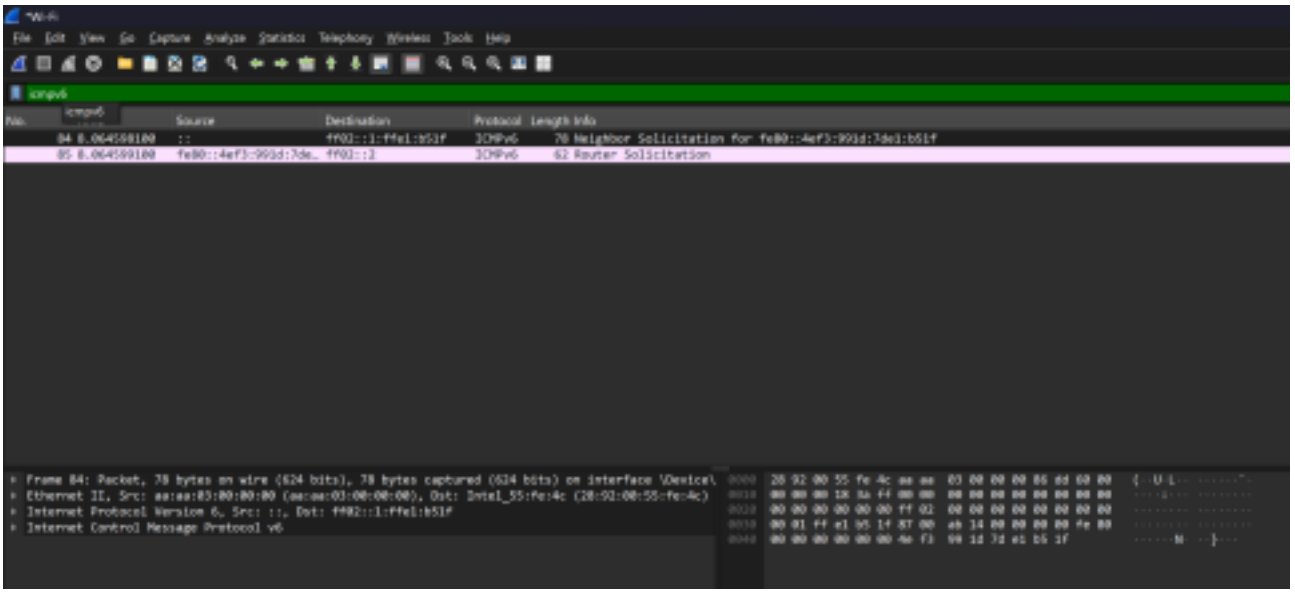
Step 4: Apply the UDP display filter

The display filter udp is applied. The selected packet is Frame 6, 86 bytes captured. The details pane shows IPv4 source 192.168.1.3, destination 224.0.0.251, UDP source port 52375 and destination port 5353.



Step 5: Apply the ICMP filter / inspect ICMPv6

The last screenshot actually uses the display filter icmpv6, not icmp. It shows ICMPv6 Neighbor Solicitation and Router Solicitation packets. The selected packet is Frame 84 and has a frame length of 78 bytes.



Step 6: TCP packet header values

Parameter asked in lab PDF	Value from your screenshot
Frame Length	78 bytes
Ethernet Header Length	14 bytes
Source MAC Address	64:fb:92:8b:31:24
Destination MAC Address	28:92:00:55:fe:4c
IPv4 Source Address	104.18.39.21
IPv4 Destination Address	192.168.1.28
IPv4 Header Length	20 bytes (IHL = 5 × 4)
IPv4 Total Length	64 bytes (0x0040 in packet bytes)
TCP Header Length	20 bytes (data offset = 5 × 4)
TCP Segment Length	44 bytes (IPv4 total length 64 − IPv4 header 20)
TCP Payload Length	24 bytes (shown as TCP Len: 24)
TCP Source Port	443
TCP Destination Port	65362

TCP length check: IPv4 total length 64 bytes = IPv4 header 20 bytes + TCP segment 44 bytes. TCP segment 44 bytes = TCP header 20 bytes + TCP payload 24 bytes.

Frame length check: Ethernet header 14 bytes + IPv4 total length 64 bytes = 78 bytes. The screenshot does not show an Ethernet FCS in the captured frame length.

Step 7: UDP packet header values

Parameter asked in lab PDF	Value from your screenshot
Frame Length	86 bytes
Ethernet Header Length	14 bytes
Source MAC Address	aa:aa:03:00:00:00 (as displayed by Wireshark)
Destination MAC Address	28:92:00:55:fe:4c
IPv4 Source Address	192.168.1.3
IPv4 Destination Address	224.0.0.251
IPv4 Header Length	20 bytes (IHL = 5 × 4)
IPv4 Total Length	72 bytes (0x0048 in packet bytes)
UDP Header Length	8 bytes
UDP Length / Datagram Length	52 bytes (0x0034 in packet bytes)
UDP Payload Length	44 bytes (UDP length 52 − UDP header 8)

UDP Source Port	52375
UDP Destination Port	5353

UDP length check: IPv4 total length 72 bytes = IPv4 header 20 bytes + UDP datagram 52 bytes. UDP datagram 52 bytes = UDP header 8 bytes + UDP payload 44 bytes.

Frame length check: Ethernet header 14 bytes + IPv4 total length 72 bytes = 86 bytes. The source MAC is reproduced exactly as shown in the screenshot; it looks unusual for a conventional Ethernet MAC address, so verify it in the expanded Ethernet II details if your instructor requires it.

Step 8: Observation table

Parameter	**TCP Packet**	**UDP Packet**
Frame Length	78 bytes	86 bytes
Ethernet Header Length	14 bytes	14 bytes
IP Header Length	20 bytes	20 bytes
Transport Header Length	20 bytes (TCP)	8 bytes (UDP)
IP Total Length	64 bytes	72 bytes
Transport Segment / Datagram Length	44 bytes (TCP segment)	52 bytes (UDP datagram)
Payload Length	24 bytes	44 bytes

Note about the lab sheet: Its Step 7 table repeats “Transport Header Length” in one row. To make the observation unambiguous, the second transport-related row above is recorded as TCP segment length / UDP datagram length. Payload length is shown separately.

Step 9: Header overhead calculation

Component	**TCP packet**	**UDP packet**
Ethernet header	14 bytes	14 bytes
IPv4 header	20 bytes	20 bytes
Transport header	20 bytes	8 bytes
Application payload	24 bytes	44 bytes
Total captured frame length	78 bytes	86 bytes
Total header overhead (Ethernet + IP + transport)	54 bytes	42 bytes
Payload efficiency	24 / 78 × 100 = 30.77%	44 / 86 × 100 = 51.16%

These efficiencies use the captured frame length shown by Wireshark and exclude an FCS because it is not present in the displayed captured lengths. The lab manual's separate theoretical example (1500-byte application data and FCS

included) gives 58 bytes overhead and 96.28% efficiency; that is a worked example, not the measured result from these screenshots.

Step 10: Answers to practical questions

1. What is encapsulation?

Encapsulation is the process of adding protocol headers to data as it moves through the network layers before transmission.

2. What is the purpose of the Ethernet header?

It supports delivery on a local network and contains the source MAC address, destination MAC address and type/length information.

3. What information is stored in the IP header?

It contains the IP version, header length, total length, source and destination IP addresses, TTL and protocol information.

4. What is the difference between TCP and UDP headers?

TCP has a minimum 20-byte header and includes sequence and acknowledgment information. UDP has an 8-byte header containing source port, destination port, length and checksum. TCP provides reliable connection-oriented delivery, while UDP has lower overhead and does not guarantee delivery.

5. Why does each protocol add a header?

Each protocol adds control information needed for its task, such as addressing, routing, protocol identification and delivery control.

6. What is packet overhead?

Packet overhead is the extra bytes used by protocol headers and trailers rather than application data.

7. How does header overhead affect network efficiency?

Headers use part of the packet for control information instead of application data. A larger overhead relative to payload reduces the proportion of the packet carrying useful data.

Conclusion

The supplied Wireshark screenshots show a successful capture on the Wi-Fi interface and the use of TCP, UDP and ICMPv6 display filters. From the selected TCP packet, the frame length is 78 bytes, IPv4 total length is 64 bytes and TCP payload is 24 bytes. From the selected UDP packet, the frame length is 86 bytes, IPv4 total length is 72 bytes and UDP payload is 44 bytes. These observations demonstrate how Ethernet, IP and transport headers contribute to packet size.